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CAMEROON GENERAL CERTIFICATE OF EDUCATION BOARD

General Certificate Of Education Examination

715 CHEMISTRY 1

JUNE 2013

ADVANCED LEVEL

Centre No. & Centre Name	
Candidate No.	
Candidate Name	

715 CHEMISTRY 1: MULTIPLE CHOICE QUESTION PAPER

One hour and a half (1H 30 mins)

INSTRUCTIONS TO CANDIDATES

Read the following instructions carefully before you start answering the questions in this paper. Make sure you have a soft HB pencil and an eraser for this examination.

1. USE A SOFT HB PENCIL THROUGHOUT THE EXAMINATION.
2. DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

Before the examination begins:

3. Check that this question booklet is headed "715 CHEMISTRY 1 - Advanced Level"
4. Insert the information required in the spaces above.
5. Insert the information required in the spaces provided on the answer sheet using your HB pencil: **Candidate Name, Centre Number and Name, Candidate Number, Subject Code and Paper Number.** Take care that you do not crease or fold the answer sheet or make any marks on it other than those asked for in these instruction.

How to answer the questions in this Examination

6. Answer **ALL** the **50** questions in this Examination. All questions carry equal marks.
7. Non-programmable calculators are allowed.
8. Each question has FOUR suggested answers: **A, B, C** and **D**. Decide on which answer is correct. Find the number of the question on the Answer Sheet and draw a horizontal line across the letter to join the square brackets for the answer you have chosen.

   For example, if **C** is your correct answer, mark **C** as shown below:

   [A] [B] ~~[C]~~ [D]

9. Mark only one answer for each question. If you mark more than one answer, you will score a zero for that question. If you change your mind about an answer, erase the first mark carefully, then mark your new answer.
10. Avoid spending too much time on any one question. If you find a question difficult, move on to the next question. You can come back to this question later.
11. Do all rough work in this booklet, using, where necessary, the blank spaces in the question booklet.
12. Mobile phones are NOT allowed in the examination room.
13. **You must not take this booklet out of the examination room. All question booklets and answer sheets will be collected at the end of the examination.**

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Questions 1 – 24 (Twenty-four questions).

Directions: Each of the questions or incomplete statements in this section is followed by four suggested answers. Select the best answer in each case.

1. Identify the type(s) of bonding found in ammonium chloride, NH_4Cl.
   - A Ionic, covalent
   - B Ionic, covalent, dative covalent
   - C Covalent, dative covalent
   - D Ionic, dative covalent

2. Which one of the following equations represents the electron affinity of chlorine?
   - A $\frac{1}{2}Cl_2(g) + \acute{e} \rightarrow Cl^-(g)$
   - B $\frac{1}{2}Cl_2(g) + \acute{e} \rightarrow Cl^-(aq)$
   - C $Cl(g) + \acute{e} \rightarrow Cl^-(g)$
   - D $Cl(g) + \acute{e} \rightarrow Cl^-(aq)$

3. An element which shows a diagonal relationship with beryllium is
   - A Oxygen
   - B Magnesium
   - C Boron
   - D Aluminium

4. The position of the equilibrium for the reaction $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$, ΔH is negative, will shift to the right if
   - A $I_2(g)$ is removed
   - B Temperature is increased
   - C Volume of the container is doubled
   - D $H_2(g)$ is added

5. In a reaction, the concentration of a reactant is varied from 0.1 to 0.2 units and the rate is found to change from 8.0×10^{-4} to 1.6×10^{-3}, all the other concentrations remaining unchanged. The order of the reaction with respect to the reactant is
   - A 2
   - B 3
   - C 1
   - D 0

6. The enthalpy of neutralization for a strong acid and a strong base is -57.3 kJ mol^{-1}. Select an equation which illustrates this.
   - A $KOH + HCl \rightarrow KCl + H_2O$
   - B $KOH + CH_3COOH \rightarrow CH_3COOK + H_2O$
   - C $2KOH + H_2SO_4 \rightarrow K_2SO_4 + 2H_2O$
   - D $NaOH + HF \rightarrow NaF + H_2O$

7. In the reaction scheme

$$CH_2CH_2 + HCl \xrightarrow[\text{Step 1}]{} CH_3CH_2Cl \xrightarrow[\text{Step 2}]{OH^-(aq),\ \text{reflux}} CH_3CH_2OH$$

The percentage conversion for steps 1 and 2 are respectively 72% and 25%. The overall percentage conversion of CH_2CH_2 to CH_3CH_2OH is:

   - A 18
   - B 97
   - C 47
   - D 48.5

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8. Which of the following pairs of liquids when mixed would give a rise in temperature?
   A Propane and tetrachlomethane
   B Propanone and hydrogen chloride
   C Ethanol and water
   D Trichloromethane and water

9. Which of the following statements about the Group VII (Group 17) elements (fluorine to iodine) is NOT correct?
   A Electronegativity decreases with increasing atomic number
   B They all show variable oxidation states in their compounds
   C They all form acidic hydrides
   D Standard electrode potentials become less positive with increasing atomic number

10. Which property of the Group IV (Group 14) elements (C – Pb) decreases with increasing atomic number?
   A The reducing power of the +2 compounds
   B The stability of the +2 compounds
   C The stability of the tetrahydrides
   D The inert pair effect

11. Barium hydroxide, $Ba(OH)_2$ of mass 8.55 g was dissolved in water and made up to 1 dm^3 of solution. What is the concentration (mol dm^{-3}) of the solution with respect to the hydroxide ion?
[Molar mass $Ba(OH)_2$ = 171]
   A 0.010
   B 0.025
   C 0.050
   D 0.100

12. A compound X with molecular formula, C_3H_8O, can be oxidized to a compound Y with molecular formula $C_3H_6O_2$. X is most likely to be
   A a primary alcohol
   B a secondary alcohol
   C an aldehyde
   D a ketone

13. The atomic number of manganese is 25. Which one of the following is the correct electronic configuration of the Mn^{2+} ion?
   A $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^2$
   B $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^0$
   C $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^2$
   D $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4 4s^1$

14. Iodoethanoic acid ionizes as follows
$CH_2ICO_2H \rightleftharpoons CH_2ICO_2^- + H^+$
The concentration of the hydrogen ions in mol dm^{-3} in a 0.1 mol dm^{-3} solution of iodoethanoic acid ($Ka = 1.0 \times 10^{-3}$ mol dm^{-3} at 25 °C) is approximately?
   A 1×10^{-1}
   B 1×10^{-2}
   C 1×10^{-3}
   D 1×10^{-4}

15. Which of the following ions results from the action of hydrochloric acid on n- butylamine (1-aminobutane)?
   A $C_4H_7NH_2^+$
   B $C_4H_9NH_3^+$
   C $C_4H_9NH_2^+$
   D $C_4H_7NH_3^+$

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16. Below are some standard electrode potentials

	E^{θ}/Volts
$Cd^{2+}(aq) / Cd(s)$	– 0.40
$Cu^{2+}(aq) / Cu(s)$	+ 0.34
$I_2(aq), 2I^-(aq)/Pt$	+ 0.54

It would be expected that

A The calculated emf of the cell $Cu(s) / Cu^{2+}(aq) \,\|\, I_2(s), 2I^-(aq)/Pt$ would be + 0.88V.
B Potassium iodide solution would reduce cadmium sulphate to cadmium
C Cadmium metal would liberate copper from aqueous copper sulphate
D The calculated emf of the cell $Cu(s) / Cu^{2+}(aq) \,\|\, Cd^{2+}(aq) / Cd(s)$ would be + 0.74V

17. Which of the following compounds will give a silver mirror when warmed with ammoniacal silver nitrate solution?

A $CH_3CH(OH)CHO$
B CH_3CH_2OH
C CH_3COCH_3
D $CH_3CH(OH)CH_3$

18. The pH value of a 0.05 mol dm^{-3} sulphuric acid is most likely to be

A 1
B 0.05
C 0.50
D 2

19. The half-life of the radio-isotope iodine – 131 is 8 days. What fraction of the original iodine – 131 (^{131}I) will remain after 24 days.

A $1/24$
B $1/8$
C $1/6$
D $1/3$

20. A 0.30 g sample of a volatile organic liquid when vapourised at s.t.p occupies 0.112 dm^3. What is the molecular formula of the liquid? (R.A.M: H = 1; C = 12; O = 16; 1 mole of any gas at s.t.p occupies 22.4 dm^3)

A CH_3OH
B CH_3COCH_3
C CH_3CHO
D $CH_3CH_2CH_2OH$

21. Which of the following compounds exhibits cis-trans (geometric) isomerism?

A $CH_2 = CHCO_2H$
B $CH_2 = CHCH_3$
C $CH_3CH = CH_2$
D $HO_2CCH = CHCO_2H$

22. Propanamide ($CH_3CH_2CONH_2$) could be prepared by

A reacting ammonia with $CH_3CH_2CO_2CH_3$
B reacting ammonia with $CH_3CO_2C_2H_5$
C adding aqueous ammonia to $CH_3CH_2CH_2Cl$
D adding CH_3COCl to $CH_3CH_2NH_2$

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23. The lattice energy of $MgCl_2$ is numerically equal to the energy change for the reaction.
   A $Mg^{2+}(s) + 2Cl^-(g) \rightarrow MgCl_2(s)$
   B $Mg^{2+}(aq) + 2Cl^-(aq) \rightarrow MgCl_2(s)$
   C $Mg^{2+}(g) + 2Cl^-(g) \rightarrow MgCl_2(s)$
   D $Mg^{2+}(g) + 2Cl^-(g) \rightarrow MgCl_2(g)$

24. Which of the following statements is NOT true for sulphur dioxide?
   A It is a reducing agent
   B It causes acid rain
   C It is an anti-oxidant
   D It is an oxidizing agent

Questions 25 – 36 (Twelve questions).
Directions: Each group of questions below consists of four lettered headings followed by a list of numbered questions. For each numbered question, select the heading which is most closely related to it. Each heading may be used once, more than once or not all.

Questions 25 – 27 concern the following types of bonding which can exist between particles.
   A Hydrogen bonding
   B Dipole – dipole interactions
   C Van der Waals forces
   D Dative covalent bonding

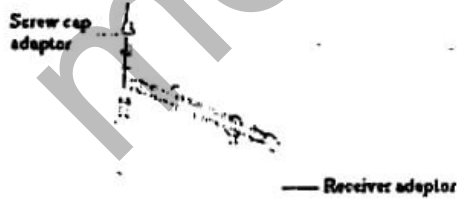
Select from A to D the type of bonding which best accounts for each of the following:

25. The melting points of the halogens increase as the group is descended.
   A
   B
   C
   D

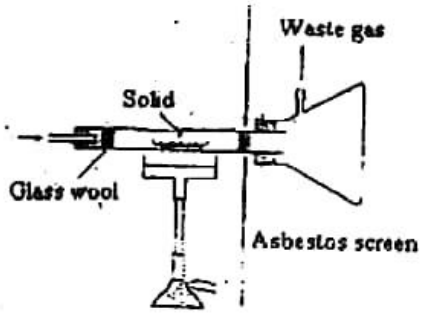
26. Ammonia reacts with BF_3
   A
   B
   C
   D

27. The addition of a little water lowers the boiling point of ethanol.
   A
   B
   C
   D

Questions 28 – 30 concern the following diagrams of experimental set ups used in some laboratory processes.



A



B

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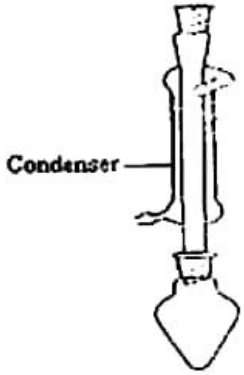
C



D

Select from A to D,

28. The diagram that would be the most suitable set up for the preparation of an ester.
A
B
C
D

29. The diagram that represents the set up for the separation of a mixture of propan-1-ol and propan -2-ol.
A
B
C
D

30. The diagram that could be used to obtain a pure sample of solid aluminium chloride
A
B
C
D

Questions 31 – 33 concern the following compounds

A $C_6H_5NH_2$
B NH_3
C $C_2H_5NH_2$
D CH_3CONH_2

Select from A to D the compound which:

31. is most basic
A
B
C
D

32. forms a stable diazonium ion at 0 °C
A
B
C
D

33. reacts with bromine and aqueous sodium hydroxide to form a gaseous compound
A
B
C
D

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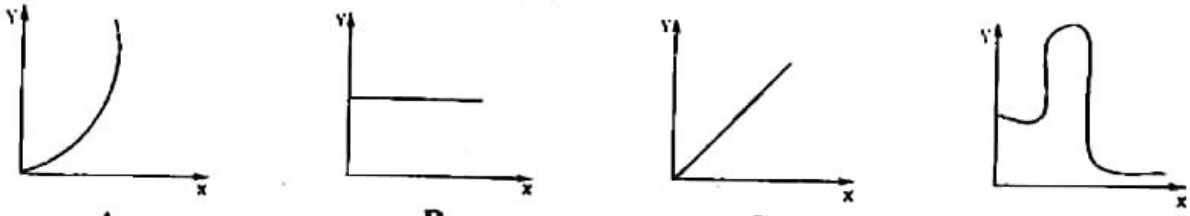
Questions 34 – 36 concern the following types of graphs:



A B C D

Select from A to D the graph that best represents.

34. A reaction with the rate equation, Rate = k[reactant]°
A
B
C
D

35. An energy profile for an exothermic reaction.
A
B
C
D

36. A first order reaction, Rate = k[reactant]
A
B
C
D

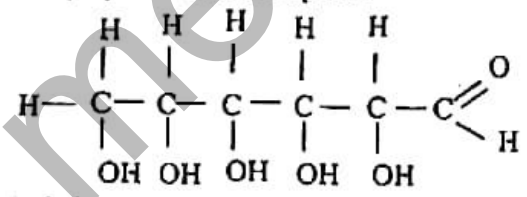
Questions 37 – 46 (Ten questions)

Directions: For each of the questions below, ONE or MORE of the responses is (are) correct. Decide which of the responses is (are) correct. Then choose:

A if 1, 2 and 3 are all correct
B if 1 and 2 only are correct
C if 2 and 3 only are correct
D if 3 only is correct

Directions Summarised			
A	B	C	D
1, 2, 3 correct	1, 2 only	2, 3 only	3 only

37. The properties of the compound



include:

1 It has 4 asymmetric carbon atoms
2 It can react with ammoniacal silver nitrate (Tollen's reagent) to give a silver mirror
3 It has only one asymmetric carbon atom

A
B
C
D

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38. The reaction

$$\underset{pink}{[Co(H_2O)_6]^{2+}(aq)} + 4Cl^-(aq) \rightleftharpoons \underset{blue}{[CoCl_4]^{2-}(aq)} + 6H_2O(l)$$

is exothermic in the forward direction. Which of the following changes to an equilibrium mixture will produce a predominantly pink solution?

1 Cooling the solution
2 The addition of concentrated HCl
3 The addition of aqueous silver nitrate

A
B
C
D

39. In which of the following complex ions is the ligand bidentate?

1 $[Cr(NH_3)_6]^{3+}$
2 $[Co(C_2O_4)_3]^{3-}$
3 $[Ni(H_2NCH_2CH_2NH_2)_2]^{2+}$

A
B
C
D

40. Hydrogen peroxide oxidizes hydroiodic acid according to the equation

$$H_2O_2(aq) + 2H^+(aq) + 2I^-(aq) \rightarrow I_2(aq) + 2H_2O(l)$$

The rate of the reaction could be investigated by measuring changes in the

1 pH
2 intensity of colour
3 electrical conductivity

A
B
C
D

41. Which of the following properties of Group I elements decrease(s) with increasing atomic number?

1 Melting temperature
2 First ionization energy
3 Atomic radius

A
B
C
D

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2. In the nuclear reaction

$$^{43}_{20}Ca + X \rightarrow ^{46}_{21}Sc + Y$$

The particles X and Y could be respectively

1 $^{2}_{1}H$ and $^{0}_{-1}e$
2 $^{4}_{2}He$ and $^{1}_{1}H$
3 $^{7}_{3}Li$ and $^{4}_{2}He$

A
B
C
D

43. Which of the following processes is (are) exothermic?

1 $O^{-}(g) + e^{-} \rightarrow O^{2-}(g)$
2 $O(g) \rightarrow O^{+}(g) + e^{-}$
3 $O(g) + e^{-} \rightarrow O^{-}(g)$

A
B
C
D

44. From the data

$$Cr^{3+}_{(aq)} + 3e^{-} \rightleftharpoons Cr_{(s)} \quad E^{\theta} = -0.74V$$
$$Fe^{3+}_{(aq)} + e^{-} \rightleftharpoons Fe^{2+}_{(aq)} \quad E^{\theta} = +0.77V$$

It could be deduced that

1 The standard e.m.f of the cell
$Cr(s) / Cr^{3+}(aq) \,||\, Fe^{3+}(aq), Fe^{2+}(aq) / Pt$, is $+1.51V$
2 $Cr(s)$ is a more powerful reductant than $Fe^{2+}(aq)$ ion
3 $Fe^{3+}(aq)$ will oxidize $Cr^{3+}(aq)$ ions under standard conditions

A
B
C
D

45. An organic liquid X is optically active and reacts with water to form an acidic solution. On addition of aqueous silver nitrate to the solution, a white precipitate forms.
The formula of X could be

1 $Cl\,CH_2CH_2COCl$
2 $Cl\,CH_2CH(CH_3)\,CHO$
3 $CH_3CHClCOCl$

A
B
C
D

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46. Propan-1-ol ($CH_3CH_2CH_2OH$) and ethanol (CH_3CH_2OH) would differ in their reactions with:

1 PCl_5
2 Metallic Na
3 Iodine in aqueous KOH

A
B
C
D

Questions 47 – 50 (Four questions)

Directions: Each of the following questions consists of a statement in the left-hand column followed by a second statement in the right-hand column. Decide whether the first statement is true or false. Decide whether the second statement is true or false. Then choose:

A If both statements are true and the second statement is a CORRECT explanation of the first statement.
B If both statements are true and the second statement is NOT a CORRECT explanation of the first statement.
C If the first statement is true, but the second statement is false.
D If the first statement is false, but the second statement is true.

Summary of Directions			
	First Statement	Second Statement	
A	True	True	Second statement is a CORRECT explanation of the first
B	True	True	Second statement is NOT a CORRECT explanation of the first
C	True	False	
D	False	True	

	FIRST STATEMENT	SECOND STATEMENT
47.	The reaction between propene (CH_3CHCH_2) and hydrogen bromide produces 2 – bromopropane ($CH_3CHBrCH_3$) as a major product and 1- bromopropane ($CH_3CH_2CH_2Br$) as a minor product.	The secondary carbocation is more stable than the primary carbocation.
48.	Hydrogen chloride has a lower boiling point than hydrogen flouride	The intermolecular forces in HCl are weaker than those in HF
49.	CH_3CO_2Na forms an acidic solution when dissolved in water	$CH_3CO^-_2$ ions in aqueous solution can combine with H^+ ions to form CH_3CO_2H molecules
50.	The process $O^-(g) + e \rightarrow O^{2-}(g)$ is endothermic	Heat is always absorbed when an atom gains an electron.

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